Python Data Structures & NumPy Scientific Computing

Python â¢ Unit 1: Python Programming â¢ Study Notes

Python data structures, functional primitives, ndarray operations, and broadcasting.

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1. Core Python Data Structures & Paradigms

Lists, tuples, dictionaries, and sets: time complexity and usage patterns.

List/dict comprehensions and generator expressions for memory efficiency.

Lambda functions, map, filter, zip, and functools.reduce in data processing.

2. NumPy Numerical Computing

N-dimensional array (ndarray) creation, striding, slicing, and memory views.

Broadcasting rules: shape compatibility across singleton dimensions.

Vectorized operations, linear algebra (np.linalg), and eigenvalue decomposition.